

VEDA Technology Internship — Task 9 Report

Image Gallery Application Documentation

1. Project Overview

This report documents the design, implementation, and empirical verification of **Task 9 (Image Gallery)** for the VEDA Technology Web Development Internship. The application delivers an interactive, modern, and accessible image gallery built strictly using HTML5, CSS3 (CSS Grid), and Vanilla JavaScript without relying on external frameworks or third-party libraries.

2. Technical Architecture & Component Stack

Component	Implementation & Technical Details
HTML5 Structure	Semantic HTML5 elements (<header>, <main>, <article>, <nav>, <footer>) with accessibility attributes (aria-hidden, aria-label, tabindex).
CSS3 Grid & Layout	Clean light theme palette (#F7F9FC background, #5577D9 accent, #263248 text). Responsive CSS Grid with breakpoints for 4-col desktop, 3-col laptop, 2-col tablet, and 1-col mobile layout.
Vanilla JS (ES6+)	State-driven lightbox preview modal, image counter, category filtering, escape key modal closing, left/right arrow keyboard navigation, and focus restoration.

3. Key Features & Technical Capabilities

Feature Name	Technical Specification	User Experience Impact
9 Curated Images	High-res photography via reliable Unsplash CDN URLs with alt text.	Ensures non-broken, visually compelling nature gallery content.
CSS Grid Layout	Fluid grid using grid-template-columns & media queries.	Adapts seamlessly from 4 columns on desktop down to 1 col on mobile.
Lightbox Modal	Full-screen backdrop with blurred overlay & smooth scale transform.	Delivers focus on enlarged image previews with captions.
Keyboard Nav	Handles Escape to close, ArrowLeft / ArrowRight to navigate.	Enables effortless accessible browsing without touching a mouse.
Category Filter	Dynamic JS filtering by category tags (Mountains, Forest, Water).	Allows instant sorting and grouping of gallery items.

4. Aspect Ratio & Accessibility Safety Compliance

- **Aspect Ratio Maintenance:** Gallery cards enforce aspect-ratio: 4 / 3 with object-fit: cover; lightbox previews enforce object-fit: contain with max-height: 75vh ensuring 0 image distortion.
- **Zero Framework Dependency:** 100% written in native HTML5, CSS3, and ES6 JavaScript.
- **Keyboard & ARIA Accessibility:** All cards are key-focusable (tabindex="0"); modal uses role="dialog" and aria-modal="true" with automatic focus lock and return.
- **Scroll Lock Prevention:** Lightbox modal dynamically locks body scroll (overflow: hidden) while active and restores scroll on close.

5. Verification Test Suite Matrix

Test Case	Execution Path	Expected Outcome	Status
Gallery Load	Page DOMContentLoaded	9 high-res images render with alt text	PASSED
Card Click Preview	Click any gallery item	Lightbox opens correct full image & title	PASSED
Close Button	Click #lightbox-close	Lightbox closes, scroll restored	PASSED

Escape Key Close	Press Escape key	Active lightbox modal closes immediately	PASSED
Arrow Key Nav	Press Left / Right arrows	Cycles smoothly to prev/next image	PASSED
Console Audit	Browser DevTools check	0 JavaScript console errors	PASSED
Mobile Layout	Viewport width < 540px	Responsive 1-col layout, zero overflow	PASSED

6. Repository & Local Verification Information

- **GitHub Repository:** <https://github.com/dharmarayank946/Image-Gallery.git>
- **Target Folder:** Task9-Image-Gallery/
- **Local Web Server:** <http://localhost:8009>